

Lecture Scribe CSE400

L7_S2_A_Revised.pdf

PDF

Please generate a lecture scribe that serves as a high-density revision tool. As you write, I want you to explicitly reason through the content step-by-step. This means you must include the reasoning directly in the scribe, explaining every definition and assumption clearly. If the material contains any proofs, derivations, or example solutions, walk through them exactly as they were presented in the context. Please make the logical flow highly explicit so that I can easily follow the sequence of thoughts during my final revision. It is critical that you do not introduce any outside material, new examples, or alternative explanations; limit your reasoning and content solely to what is present in the provided text. My goal is a concise, logical map of the actual lecture that I can refer to just before my exam.

Below is a **high-density, step-by-step lecture scribe** based strictly on the uploaded lecture slides
CSE400 – Lecture 7: Expectation, CDFs, PDFs and Problem Solving

L7_S2_A_Revised

CSE400 – Lecture 7

Expectation, CDFs, PDFs and Problem Solving

(January 27, 2025 – Dhaval Patel, PhD)

1. Lecture Scope and Structure

According to the outline, the lecture proceeds in the following order :

L7_S2_A_Revised

1. Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Examples
2. Probability Density Function (PDF)
 - Definition
 - Relationship with CDF
3. Expectation of Random Variables (introduced later)

This scribe follows the same sequence.

2. Intuition: CDF and PDF (Water Tank Analogy)

Water Tank Model

From pages 3–4 :
L7_S2_A_Revised

- Consider a water tank of height **H** and radius **R**.
- Let **h** be the water height.

Interpretation

1. The height **h** represents the value of a random variable.
2. The volume of water up to height **h**, denoted **V(h)**, represents cumulative probability.

So:

- Height → random variable value
- Volume → cumulative probability

Volume Calculation

From the slide:

$$V(h) = \int_0^h \pi R^2 dh = (\pi R^2)h$$

Step-by-step meaning:

1. Cross-sectional area = πR^2 .
2. Volume = Area × Height.
3. Integrate area from 0 to **h** → $(\pi R^2)h$.

Analogy with Probability

- πR^2 corresponds to a **uniform PDF**.
- Maximum volume:

$$V(H) = \pi R^2 H$$

This corresponds to total probability = 1.

Thus:

- CDF = accumulated “volume”
- PDF = “density” of volume per unit height

3. Cumulative Distribution Function (CDF)

3.1 Definition

From pages 5–6 :
L7_S2_A_Revised

For a random variable X ,

$$F_X(x) = Pr(X \leq x), \quad -\infty < x < \infty$$

Step-by-step meaning:

1. Choose a value x .
2. Consider the event " $X \leq x$ ".
3. Compute its probability.
4. That probability is the CDF at x .

So, CDF measures how much probability has accumulated up to x .

3.2 Importance

From page 6:

Most information about the experiment is determined by the behavior of $F_X(x)$.

Meaning:

- If we know the full CDF, we can derive probabilities.
 - The CDF completely characterizes the random variable.
-

3.3 Properties of CDF

From pages 8–11 :
L7_S2_A_Revised

Property 1: Range

$$0 \leq F_X(x) \leq 1$$

Reasoning:

- $F_X(x)$ is a probability.
 - All probabilities lie between 0 and 1.
-

Property 2: Limits

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Reasoning:

1. At $-\infty$:
 $\Pr(X \leq \text{very small number}) = 0.$
2. At $+\infty$:
 $\Pr(X \leq \text{very large number}) = 1.$

Property 3: Monotonicity

For $x_1 < x_2$:

$$F_X(x_1) \leq F_X(x_2)$$

Reasoning:

1. Event $\{X \leq x_1\}$ is contained in $\{X \leq x_2\}$.
2. Larger set $\rightarrow$ larger probability.
3. So CDF is non-decreasing.

Property 4: Interval Probability

For $x_1 < x_2$:

$$\Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

Derivation:

1. $\{X \leq x_2\} = \{X \leq x_1\} \cup \{x_1 < X \leq x_2\}$
2. These events are disjoint.
3. So:

$$\Pr(X \leq x_2) = \Pr(X \leq x_1) + \Pr(x_1 < X \leq x_2)$$

4. Rearranging gives the formula.

4. CDF: Example #1 (Validity Check)

From pages 12-15 :
L7_S2_A_Revised

We are asked to find valid CDFs.

(a) $F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$

Check step-by-step:

1. Range: $\arctan(x) \in (-\pi/2, \pi/2)$
2. So:

$$\frac{1}{\pi} \tan^{-1}(x) \in \left(-\frac{1}{2}, \frac{1}{2}\right)$$

3. Adding 1/2 shifts range to (0,1).

4. Limits:

- $x \rightarrow -\infty \Rightarrow \arctan \rightarrow -\pi/2 \Rightarrow F \rightarrow 0$
- $x \rightarrow +\infty \Rightarrow \arctan \rightarrow \pi/2 \Rightarrow F \rightarrow 1$

5. Function is increasing.

✓ Valid CDF

(b) $F_X(x) = (1 - e^{-x})u(x)$

Here $u(x)$ is unit step.

Meaning:

- For $x < 0$: $F = 0$
- For $x \geq 0$: $F = 1 - e^{-x}$

Check:

1. At $x < 0$: probability = 0
2. At $x \rightarrow \infty$: $e^{-x} \rightarrow 0 \Rightarrow F \rightarrow 1$
3. Function is increasing.

✓ Valid CDF

(c) $F_X(x) = e^{-x^2}$

Check:

1. At $x \rightarrow -\infty$: $e^{-x^2} \rightarrow 0$
2. At $x \rightarrow +\infty$: $e^{-x^2} \rightarrow 0$

Limit is not 1.

✗ Invalid (violates $F(\infty)=1$)

(d) $F_X(x) = x^2 u(x)$

For $x \geq 0$: $F = x^2$

As $x \rightarrow \infty$, $F \rightarrow \infty$, not 1.

✗ Invalid

5. CDF: Example #2 (Probability Computation)

From page 16 :
L7_S2_A_Revised

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) $Pr(X > 5)$

Step-by-step:

$$\begin{aligned} Pr(X > 5) &= 1 - Pr(X \leq 5) = 1 - F_X(5) \\ &= 1 - (1 - e^{-5}) = e^{-5} \end{aligned}$$

(b) $Pr(X < 5)$

$$Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) $Pr(3 < X < 7)$

$$\begin{aligned} &= F_X(7) - F_X(3) \\ &= (1 - e^{-7}) - (1 - e^{-3}) \\ &= e^{-3} - e^{-7} \end{aligned}$$

(d) $Pr(X > 5 \mid X < 7)$

Conditional probability:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Let:

- $A = \{X > 5\}$
- $B = \{X < 7\}$

Then:

$$A \cap B = \{5 < X < 7\}$$

So:

$$= \frac{F(7) - F(5)}{F(7)}$$

6. Probability Density Function (PDF)

6.1 Definition

From page 17 :
L7_S2_A_Revised

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

Meaning:

1. Take a very small interval of width ε .
2. Compute probability inside it.
3. Divide by width.
4. Take limit.

This gives “probability per unit length”.

6.2 Using CDF

From page 18:

$$Pr(x < X < x + \varepsilon) = F_X(x + \varepsilon) - F_X(x)$$

Substitute:

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_X(x + \varepsilon) - F_X(x)}{\varepsilon}$$

6.3 PDF-CDF Relationship

From page 19:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Reasoning:

1. The limit above is the definition of derivative.
 2. Therefore, PDF is derivative of CDF.
-

6.4 Inverse Relationship

From pages 20–21 :
L7_S2_A_Revised

PDF is the derivative of CDF.

CDF is the integral of PDF.

So:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Graphically:

- Area under PDF curve up to x = CDF value.

7. Expectation of Random Variables (Intro)

Pages 33–35 introduce “Expectation of RVs” L7_S2_A_Revised, but no detailed formulas are visible in the extracted content.

Thus, in this file segment:

- Only the heading “Expectation of RVs” is introduced.
- Detailed derivations are not present here.

So, no further steps are included in this section from the provided material.

✓ Final Logical Map (Exam Revision)

1. CDF

$$F_X(x) = Pr(X \leq x)$$

Properties:

- $0 \leq F \leq 1$
- $F(-\infty) = 0, F(\infty) = 1$
- Non-decreasing
- $Pr(a < X \leq b) = F(b) - F(a)$

2. Validity Check

To test a CDF:

1. Check limits ($0 \rightarrow 1$)
2. Check monotonicity
3. Check range

3. Probability from CDF

$$Pr(a < X < b) = F(b) - F(a)$$

$$Pr(X > x) = 1 - F(x)$$

4. PDF

Definition:

$$f(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

Relation:

$$f(x) = \frac{dF(x)}{dx}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

If you'd like, I can next prepare the same **step-by-step high-density scribe** for the *Expectation* part when you share those slides.

Sources

L7_S2_A_Revised.pdf

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Below is a **concise, high-density, step-by-step lecture scribe** based strictly on the uploaded slides **CSE400 – Lecture 7: Expectation, CDFs, PDFs and Problem Solving**.

L7_S2_A_Revised

All definitions, derivations, and examples are explained only using the provided material.

CSE400 – Lecture 7

Expectation, CDFs, PDFs and Problem Solving

(January 27, 2025)

1. Lecture Structure (Page 2)

According to the outline:

1. Cumulative Distribution Function (CDF)
 - Definition
 - Properties
 - Examples
2. Probability Density Function (PDF)
 - Definition
 - CDF-PDF Relationship
3. Expectation of Random Variables (introduced later)

The lecture follows this order.

2. Intuition: CDF and PDF (Water Tank Analogy)

(Pages 3–4)

Given Model

- A cylindrical water tank:
 - Radius = R
 - Height = H
- Water height = h

Interpretation

The slide states:

- Height $h \rightarrow$ value of random variable X
- Volume up to $h \rightarrow$ cumulative probability

So:

- Each height corresponds to a value of X .
- Accumulated water corresponds to $\Pr(X \leq h)$.

Thus:

$$\text{Volume up to } h \longleftrightarrow F_X(h)$$

Volume Calculation

Given:

$$V(h) = \int_0^h \pi R^2 dh$$

Step-by-step:

1. Cross-sectional area = πR^2 .
2. Volume = Area $\times$ Height.
3. Integrate area from 0 to h.
4. Result:

$$V(h) = (\pi R^2)h$$

Probability Interpretation

Maximum volume:

$$V(H) = \pi R^2 H$$

This corresponds to total probability = 1.

Thus:

- πR^2 behaves like a constant PDF.
- $V(h)$ behaves like a CDF.
- CDF = accumulated PDF.

3. Cumulative Distribution Function (CDF)

3.1 Definition

(Pages 5–6)

The slide defines:

$$F_X(x) = Pr(X \leq x), \quad -\infty < x < \infty$$

Step-by-step meaning:

1. Fix a real number x.
2. Consider the event: " $X \leq x$ ".
3. Compute its probability.
4. That probability is $F_X(x)$.

So:

The CDF gives the total probability accumulated up to x .

Importance (Page 6)

The slide states:

Most information about the random experiment is determined by the behavior of $F_X(x)$.

Meaning:

- Knowing $F_X(x)$ allows us to compute all probabilities.
 - CDF fully characterizes the random variable.
-

3.2 Properties of CDF

(Pages 8–11)

Property 1: Boundedness

$$0 \leq F_X(x) \leq 1$$

Reasoning:

- $F_X(x)$ is a probability.
 - All probabilities lie between 0 and 1.
-

Property 2: Limits

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Reasoning:

1. At $-\infty$, $\Pr(X \leq \text{very small}) = 0$.
 2. At $+\infty$, $\Pr(X \leq \text{very large}) = 1$.
-

Property 3: Monotonicity

For $x_1 < x_2$:

$$F_X(x_1) \leq F_X(x_2)$$

Reasoning:

1. $\{X \leq x_1\} \subset \{X \leq x_2\}$.
2. Larger set $\rightarrow$ larger probability.
3. Therefore, CDF is non-decreasing.

Property 4: Interval Probability

For $x_1 < x_2$:

$$Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

Derivation:

1. $\{X \leq x_2\} = \{X \leq x_1\} \cup \{x_1 < X \leq x_2\}$.

2. These sets are disjoint.

3. So:

$$Pr(X \leq x_2) = Pr(X \leq x_1) + Pr(x_1 < X \leq x_2)$$

4. Rearranging:

$$Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

4. CDF Example #1: Validity Check

(Pages 12-15)

Question: Find valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Step-by-step check:

1. Range of $\arctan(x)$: $(-\pi/2, \pi/2)$.

2. Divide by π : $(-1/2, 1/2)$.

3. Add $1/2$: $(0,1)$.

4. Limits:

- $x \rightarrow -\infty \Rightarrow F \rightarrow 0$
- $x \rightarrow \infty \Rightarrow F \rightarrow 1$

5. Function is increasing.

✓ Valid CDF

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Meaning:

- $x < 0 \Rightarrow F = 0$
- $x \geq 0 \Rightarrow F = 1 - e^{-x}$

Check:

1. $F \geq 0$
2. $F(\infty) = 1$
3. Non-decreasing

✓ Valid CDF

(c)

$$F_X(x) = e^{-x^2}$$

Check:

- $x \rightarrow \infty \Rightarrow F \rightarrow 0$, not 1

✗ Invalid (violates limit property)

(d)

$$F_X(x) = x^2 u(x)$$

For $x \geq 0$:

- $F = x^2 \rightarrow \infty$

✗ Invalid (exceeds 1)

5. CDF Example #2: Probability Computation

(Page 16)

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) $Pr(X > 5)$

$$\begin{aligned} Pr(X > 5) &= 1 - Pr(X \leq 5) \\ &= 1 - F_X(5) \\ &= 1 - (1 - e^{-5}) = e^{-5} \end{aligned}$$

(b) $Pr(X < 5)$

$$Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) $Pr(3 < X < 7)$

$$\begin{aligned} &= F_X(7) - F_X(3) \\ &= (1 - e^{-7}) - (1 - e^{-3}) \\ &= e^{-3} - e^{-7} \end{aligned}$$

(d) $Pr(X > 5 \mid X < 7)$

Using conditional probability:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Let:

- $A = \{X > 5\}$
- $B = \{X < 7\}$

Then:

$$A \cap B = \{5 < X < 7\}$$

So:

$$= \frac{F(7) - F(5)}{F(7)}$$

6. Probability Density Function (PDF)

6.1 Definition

(Page 17)

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

Step-by-step meaning:

1. Take small interval $(x, x + \varepsilon)$.
2. Find probability inside it.
3. Divide by width ε .

4. Take limit.

This gives probability per unit length.

6.2 Using CDF

(Page 18)

From slide:

$$\Pr(x < X < x + \varepsilon) = F_X(x + \varepsilon) - F_X(x)$$

Substitute into definition:

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_X(x + \varepsilon) - F_X(x)}{\varepsilon}$$

6.3 PDF-CDF Relationship

(Page 19)

The limit equals derivative:

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Reasoning:

- The limit matches the definition of derivative.
 - So PDF is derivative of CDF.
-

6.4 Inverse Relationship

(Pages 20–21)

The slide states:

- PDF = derivative of CDF
- CDF = integral of PDF

So:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Graphically:

- Area under PDF up to $x = F_X(x)$.
-

7. Expectation of Random Variables

(Pages 33–35)

In the provided file segment:

- Only the heading “Expectation of RVs” appears.
- No formulas or derivations are visible.

Therefore:

- No further content is available in this file section.
- No additional steps can be derived.

✓ Final Exam-Oriented Logical Map

CDF

$$F_X(x) = Pr(X \leq x)$$

Properties:

- $0 \leq F \leq 1$
- $F(-\infty) = 0, F(\infty) = 1$
- Non-decreasing
- $Pr(a < X \leq b) = F(b) - F(a)$

Valid CDF Check

Must satisfy:

1. Range: $[0,1]$
2. Limits: $0 \rightarrow 1$
3. Monotonicity

Probabilities from CDF

$$Pr(X > x) = 1 - F(x)$$

$$Pr(a < X < b) = F(b) - F(a)$$

PDF

$$f(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

$$f(x) = \frac{dF(x)}{dx}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

If you upload the remaining slides on **Expectation and Moments**, I can prepare the same step-by-step revision scribe for those as well.

Sources

L7_S2_A_Revised.pdf

PDF

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All reasoning, definitions, and examples follow the exact logical flow of the slides, with no external material added.

CSE400 – Lecture 7

Expectation, CDFs, PDFs and Problem Solving

(January 27, 2025)

1. Lecture Organization (Page 2)

The outline states that the lecture proceeds in this order:

1. Cumulative Distribution Function (CDF)

- Definition

- Properties
- Examples
- 2. Probability Density Function (PDF)
 - Definition
 - Relationship with CDF
- 3. Expectation of Random Variables
 - Definition and Examples
 - Functions of RVs
 - Linear Operations
 - Moments

This scribe follows the same structure.

2. Intuition: CDF and PDF (Water Tank Analogy)

(Pages 3–4)

Physical Model

- A cylindrical tank:
 - Radius = R
 - Height = H
 - Water height = h
-

Interpretation

From the slide:

- Height h represents a value of random variable X .
- Volume up to height h represents cumulative probability.

Thus:

$$h \longleftrightarrow x, \quad V(h) \longleftrightarrow F_X(x)$$

Meaning:

- Each value of X corresponds to a water level.
 - Probability up to that value corresponds to water volume.
-

Volume Computation

Given:

$$V(h) = \int_0^h \pi R^2 dh$$

Step-by-step:

1. Cross-sectional area = πR^2 .
2. Volume = area $\times$ infinitesimal height dh .
3. Integrate from 0 to h .
4. Result:

$$V(h) = (\pi R^2)h$$

Probability Meaning

Maximum volume:

$$V(H) = \pi R^2 H$$

This corresponds to total probability = 1.

So:

- πR^2 behaves like a constant PDF.
- $V(h)$ behaves like a CDF.
- CDF = accumulated PDF.

3. Cumulative Distribution Function (CDF)

3.1 Definition

(Pages 5–6)

The slide defines:

$$F_X(x) = Pr(X \leq x), \quad -\infty < x < \infty$$

Step-by-step meaning:

1. Choose a real number x .
2. Consider event $\{X \leq x\}$.
3. Compute its probability.
4. That probability is the CDF at x .

So:

CDF measures accumulated probability up to x .

Importance (Page 6)

The slide states:

Most information about the experiment is determined by $F_X(x)$.

Meaning:

- If $F_X(x)$ is known, all probabilities can be derived.
 - The random variable is completely characterized.
-

3.2 Properties of CDF

(Pages 8–11)

Property 1: Bounds

$$0 \leq F_X(x) \leq 1$$

Reasoning:

- $F_X(x)$ is a probability.
 - All probabilities lie in $[0,1]$.
-

Property 2: Limits

$$F_X(-\infty) = 0, \quad F_X(\infty) = 1$$

Reasoning:

1. Very small $x \Rightarrow$ almost no values $\leq x$.
 2. Very large $x \Rightarrow$ all values $\leq x$.
-

Property 3: Monotonicity

For $x_1 < x_2$:

$$F_X(x_1) \leq F_X(x_2)$$

Reasoning:

1. $\{X \leq x_1\} \subset \{X \leq x_2\}$.
 2. Larger event $\rightarrow$ larger probability.
 3. Hence, CDF is non-decreasing.
-

Property 4: Interval Probability

For $x_1 < x_2$:

$$Pr(x_1 < X \leq x_2) = F_X(x_2) - F_X(x_1)$$

Derivation:

1. $\{X \leq x_2\} = \{X \leq x_1\} \cup \{x_1 < X \leq x_2\}$.
 2. Disjoint events.
 3. Add probabilities.
 4. Rearrange.
-

4. CDF Example #1: Validity

(Pages 12-15)

Question: Find valid CDFs.

(a)

$$F_X(x) = \frac{1}{2} + \frac{1}{\pi} \tan^{-1}(x)$$

Check:

1. $\arctan(x) \in (-\pi/2, \pi/2)$.
2. Divide by $\pi \rightarrow (-1/2, 1/2)$.
3. Add $1/2 \rightarrow (0, 1)$.
4. Limits: $0 \rightarrow 1$.
5. Increasing.

✓ Valid

(b)

$$F_X(x) = (1 - e^{-x})u(x)$$

Meaning:

- $x < 0$: $F=0$
- $x \geq 0$: $F=1-e^{-x}$

Check:

1. Non-negative.
2. Approaches 1.
3. Increasing.

✓ Valid

(c)

$$F_X(x) = e^{-x^2}$$

Check:

- $x \rightarrow \infty \Rightarrow F \rightarrow 0$

✗ Invalid (violates $F(\infty)=1$)

(d)

$$F_X(x) = x^2 u(x)$$

Check:

- For large x , $F \rightarrow \infty$

✗ Invalid

5. CDF Example #2: Probability Calculation

(Page 16)

Given:

$$F_X(x) = (1 - e^{-x})u(x)$$

(a) $Pr(X > 5)$

$$\begin{aligned} Pr(X > 5) &= 1 - Pr(X \leq 5) \\ &= 1 - F_X(5) = e^{-5} \end{aligned}$$

(b) $Pr(X < 5)$

$$Pr(X < 5) = F_X(5) = 1 - e^{-5}$$

(c) $Pr(3 < X < 7)$

$$\begin{aligned} &= F_X(7) - F_X(3) \\ &= e^{-3} - e^{-7} \end{aligned}$$

(d) $Pr(X > 5 \mid X < 7)$

Using:

$$Pr(A|B) = \frac{Pr(A \cap B)}{Pr(B)}$$

Let:

- $A=\{X>5\}$
- $B=\{X<7\}$

Then:

$$A \cap B = \{5 < X < 7\}$$

So:

$$= \frac{F(7) - F(5)}{F(7)}$$

6. Probability Density Function (PDF)

6.1 Definition

(Page 17)

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

Meaning:

1. Consider small interval $(x, x+\varepsilon)$.
2. Compute probability.
3. Divide by width.
4. Take limit.

This gives probability density.

6.2 Using CDF

(Page 18)

$$Pr(x < X < x + \varepsilon) = F_X(x + \varepsilon) - F_X(x)$$

Substitute:

$$f_X(x) = \lim_{\varepsilon \rightarrow 0} \frac{F_X(x + \varepsilon) - F_X(x)}{\varepsilon}$$

6.3 PDF-CDF Relationship

(Page 19)

$$f_X(x) = \frac{dF_X(x)}{dx}$$

Reasoning:

- The limit equals derivative definition.

6.4 Inverse Relationship

(Pages 20–21)

From slide:

- PDF = derivative of CDF
- CDF = integral of PDF

So:

$$F_X(x) = \int_{-\infty}^x f_X(t) dt$$

Area under PDF = CDF.

7. Expectation of Random Variables

(Pages 33–35)

In the provided file:

- Only the heading “Expectation of RVs” appears.
- No formulas or examples are visible.

Therefore, no further derivations are available in this document.

Final High-Density Exam Map

CDF

$$F_X(x) = Pr(X \leq x)$$

Properties:

- $0 \leq F \leq 1$
- $F(-\infty) = 0, F(\infty) = 1$

- Non-decreasing
- $Pr(a < X \leq b) = F(b) - F(a)$

PDF

$$f(x) = \lim_{\varepsilon \rightarrow 0} \frac{Pr(x < X < x + \varepsilon)}{\varepsilon}$$

$$f(x) = \frac{dF(x)}{dx}$$

$$F(x) = \int_{-\infty}^x f(t) dt$$

CDF Validity Check

Must satisfy:

1. Range [0,1]
2. Limits $0 \rightarrow 1$
3. Monotonicity

If you upload the remaining slides on **Expectation and Moments**, I can extend this into a complete final-revision scribe for the whole unit.

Sources